

Evading and Avoiding Along Borders: How Trade Variables Affect the Size of Shadow Economies

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Abstract

Discussions regarding the shadow economy reach a wide variety of economics subjects, yet how trade variables specifically effect the size of shadow economies is understudied. This study utilizes a wide variety of trade variables and controls that make sure to avoid the mechanical collinearity problem of the MIMIC estimates while including nonlinearities and interaction terms, not present in much of the research. This study finds that tariff rates have the opposite affect of what theory would suggest. The impositon of sanctions also has large positive effects on the shadow economy, while focusing on developing comparative advantages seems a good way to reduce informality, given sufficiently small agricultural sector size. This research is important as ways to reduce informality are difficult to implement and shadow economies add strain to the government and distort statistics.

1 Introduction

The study of the shadow economy of countries is quite extensive, yet consistent results are often-times difficult to come by. There have been a number of determinants of shadow-economic size found, yet policy solutions are quite involved and difficult to put into practice. The shadow economy has a number of important policy implications and consequences. It is important to note that this phenomenon has seen some historical growth as the sizes of the shadow economies of all of the GNP nations have grown significantly in the period of 1960 - 1995 (Schneider & Enste, 2000). With an average percentage point increase of 10.4 (Schneider & Enste, 2000) and an average percentage increase of 486%, the issue of shadow economic size has been a growing consideration among researchers in the field of economics. Another startling statistic is that, according to the estimates of Medina and Schneider (2017), the largest shadow economies have their sizes estimated at around 63% of GDP (Bolivia, Guatemala, and Nigeria). This makes clear that shadow activity is a pervasive phenomenon and much must be done to address this issue. The main consequences of shadow activity can be summarized with the following.

To start, businesses that operate part of or all of their business in the shadow economy have the advantage in that they avoid paying taxes and regulatory fees (Tanzi, 2002). They can also skirt many of the regulations that “official” businesses are beholden to (Tanzi, 2002). This leads to unfair competition and can further lead businesses into the shadow economy as they try to even the playing field.

Adding to this, it is quite easy to see how a larger shadow economy would lead to a reduced tax base, thus putting a strain on social services. To compound this problem, many studies have found that the potentially heavy burden of taxes and especially social security contributions cause individuals to enter the shadow economy (Schneider & Enste, 2000). This creates a vicious cycle in which the presence of an abnormally large shadow economy causes a reduction in tax revenues, which causes a growth in the size of the shadow economy, which then puts a greater strain on the tax base and so on. It is likely that mitigating the size of the shadow economy will lead to a reduced tax burden, higher quality public services, and higher equality in taxation. We can also see the reverse of the cycle outlined above as greater incentive to tackle the issue of the shadow economy can deter shadow activity, which then reduces the strain that shadow activity produces, which can further reduce sizes.

Additionally, increased shadow economic activity can discourage foreign direct investment (FDI) (Tanzi, 2002), particularly Greenfield investments (a component of FDI characterized by a company constructing operations without the purchase of partnership of local businesses) (Yusuf & Nguyen, 2023). This, in part, likely leads to a reduction in GDP growth that is often larger than the growth rate of a country’s shadow economy (Loayza, 1996). Not only does the shadow economy have an effect on economic variables, it also distorts estimations of them. Since this activity is productive, laborious, revenue- and output-producing, and taxable, it is factored in a country’s theoretical national income statistics, but since these activities cannot be reliably estimated, the national income statistics of nearly all countries are technically inaccurate. The higher the variability and uncertainty in these measures (note that they are quite high as illustrated by Medina and Schneider (2018)), the less governments and people can rely on these as accurate representations of their society, which can obviously lead to poor policy selection among other things.

Despite the wealth of research on the topic in general, research into the relationship of trade and the shadow economy is quite fringe and emblematic of mixed and uncertain

results. Various papers have found results that are either significant or insignificant and either positive or negative. There has been no clear consensus on even the direction of this change, let alone the direction with a stable idea of significance. Theoretical implications tell us that an increase in trade liberalization will lead to increased competition, which will push producers into engaging in the shadow economy (mainly through labor as described by (Fiess & Fugazza, 2012)) for an edge. Additional theoretical work has shown that openness to trade leads to higher informal employment and higher underground wages (Kar & Marjit, 2001) (Marjit & Acharyya, 2003). Informal employment does not tell the whole story, but is a substantial part of the shadow economy and is a major motivator of its size and growth. Among empirical works, a variety of different results have been found. With similar frameworks and control variables, a clear determination of the relationship between trade and the size of the shadow economy is ambiguous.

To remedy this, a more in-depth set of econometric models need to be utilized and the relationship needs to be examined in a larger variety of ways. To start, this study uses a set of control variables that is more in-line with the seminal literature on the shadow economy in general while also eliminating the mechanical collinearity issues that are present in other papers. To truly isolate the effects of trade, it is crucial to choose a more comprehensive set of control variables. Additionally, current literature lacks diversity in the model selection. Without the presence of growth of the size of the shadow economy, certain nonlinearities, and interaction terms, it cannot be hoped to obtain a fundamental understanding of the relationship under study. Along with the preceding, certain variables that may better capture the effects of the certain types of underground activity are lacking in the literature and an analysis of these will also be present.

This paper will utilize the MIMIC model estimates to try to capture all parts of the shadow economy. Despite this less prevalent convention, it seems to be the most appropriate dataset due to its accuracy and usage across other papers (this aids in comparison). To also capture this relationship on a global scale, panel data models will be used as well as a sample of many countries across a number of years. A large breakdown will be done starting with purely linear models (to better reflect the current literature); then adding a number of control variables; and finally incorporating alterations to standard models such as interactions terms and nonlinearities. Also, endogeneity is a pervasive issue in shadow economics research, so 2sCOPE will be used to correct for this endogeneity as suitable instruments for taxation related variables are hard to come by and justify theoretically.

The results of this study have found that increasing tariff rates can help to reduce the sizes of shadow economies while all taxes on international trade work to increase shadow activity. Sanctions increase informality, while comparative advantages generally reduce informality, given a relatively average size of the agricultural sector. Foreign direct investment also increases informality. The imposition of nonlinearities do not produce significant results, thus possible gains from taxation after certain levels are achieved (concave tax functions) are not found to be important. The slope of the applied tariff rates parameter can not be assumed to be significantly different in the presence of an import sanction. Finally, endogeneity exists within this model and controlling for it via 2sCOPE alter the significance and sign of certain parameters.

2 Literature Review

2.1 Understanding the Shadow Economy

One may have noticed that, thus far, the term "shadow economy" has exclusively been used. A major issue in the study of shadow economics is that there is no globally accepted definition of it. There are a number of alternative definitions and these can be encapsulated in different terms such as black economy, dark economy, informal economy / sector, underground economy, and others. For this paper, all of the above and others not stated (but relevant) will be used synonymously. There are also various axes in which shadow economic activity can be defined, namely monetary / nonmonetary transactions and legal / illegal activities. Different estimates utilize different parts of this 2-dimensional scheme. Table 1 below, by (Schneider & Enste, 2000), presents a good summary of the various components that can be included in the definition of the informal economy.

	Monetary Transaction		Nonmonetary Transactions	
Illegal Activities	Trade in stolen goods, drug dealing and manufacturing, prostitution, gambling, smuggling, and fraud.		Barter: drugs, stolen goods, smuggling, etc. Produce or growing of drugs for own use.	
	Tax Evasion	Tax Avoidance	Tax Evasion	Tax Avoidance
Legal Activities	Unreported income from self-employment	Employee discounts	Barter of legal services and goods	All do-it-yourself work and neighbor help
	Wages, salaries and assets from unreported work related to legal services and goods	Fringe benefits.		

Figure 1: A Taxonomy of Underground Economic Activities (Schneider & Enste, 2000)

To encapsulate the relevant aspects that will be studied in this paper, while also aligning with a large portion of the research, the definition given by Schneider and Buehn (2017) will be used: "all market-based legal production of goods and services that are deliberately concealed from public authorities...". They outline the various reasons mainly to avoid paying certain taxes as well as avoiding certain regulations or administrative requirements. Generally, the "Legal Activities" part of the above table will be included under the definition of the shadow economy and all of its synonyms. This will motivate the selection of our datasets and variables to better capture all effects and implications.

2.2 Main Causes of the Shadow Economy

Among the most cited reasons for shadow economic activity are tax-related issues, the regulatory environment, and the strength of institutions. Below will also include a discussion of some other, more to-the-point determinants.

To start, the often most cited reason for the growth of the shadow economy is the prevalence of taxes. Namely, businesses who want to avoid paying taxes will transfer some of their activities into the shadow economy where they can, through tax evasion and tax avoidance, save money on tax payments. This is mainly from the business perspective, yet one can localize this to individuals, wherein, the burden of taxes can cause individuals to move to the informal economy. This happens, oftentimes, for two reasons. First of all, income taxes play a large role in this phenomenon as many, feeling the effects of a large income tax rate (i.e. a reduction in wages), will try to find work underground, where “wages” are higher (Schneider & Enste, 2000). And since this work is paid in cash and the activity illegal, the employer will not levy the tax on the individual. This is further supported by the findings of Schneider (1994) that direct taxes have the largest impact on the size of the shadow economy.

Additionally, large social security burdens will cause people to try to find a way out of them. This can realistically be done only one way, by losing your employment. When individuals are unemployed, they are not expected to pay these contributions, so they will move their labor to the shadow economy. This way, they can get paid, collect public-funded benefits, and not contribute to the society’s social safety net. It is clear why this cycle can create major issues. As was noted in the introduction, the presence of a larger shadow economy can create a cycle, wherein, the tax base is reduced, leading to a strained social services system, and thus more shadow activity, rinse and repeat.

Another important factor related to taxation is the strength of the tax-collecting institutions. Although institutions are an issue of their own, it is worth mentioning this here. It is found in a variety of studies, summarized by Schneider and Enste (2000) and Tanzi (2002), that a weaker tax-collection system emboldens individuals to act in the informal sector. Tanzi (2002) describes this as a “cold war” between tax agencies and those who are in / willing to supplement “above ground” activity with underground activity. The higher the likelihood and the greater the punishment for evading / avoiding taxes, the less individuals will want to risk entering the shadow economy. This operates as an incentive structure where people, as they often do in other things, will weigh the risks and rewards of engaging in illicit activities. Similarly, Cebula (1997) finds that more IRS audits and higher penalties imposed for mis-representing due tax payments, decreases the size of the shadow economy. Another component of institutions is the tax laws themselves, specifically the ability to legally avoid taxes. It was found by Neck et al. (2012) that the more complex a tax system (i.e. the more wiggle room for avoidance) the lower the shadow economic activity a country will have in aggregate.

The second major driver of shadow economic activity is the prevalence of regulation. It is not difficult to see that an increase in regulation, similar to taxes, will cause people to shift towards the shadow economy. Increased labor regulations can increase costs which can pressure business to supplement labor / goods purchasing in the shadows, while also passing higher prices onto consumers, thus giving them an incentive to supplement goods purchases in the shadows. This operates similar to the taxation example above and has similar, wide-reaching effects. Tanzi et al. (1983) supports this view, finding that with an overabundance of regulation comes a larger share of shadow economic activity. It is also important to note that it isn’t just the amount of regulation, but the quality of its enforcement. This is mainly related to institutions, but has its significance in regulations specifically.

The third reason that economists often cite as an explanation for shadow economic activity is the quality of institutions and, by extension, corruption. As was noted earlier, the strength of tax and regulatory institutions adds a layer of nuance to their respective

determinants. Although tax and regulation law are major factors, the ability of institutions to enforce these provisions is considerable. From this, we see a similar cycle to the one outlined in the introduction: weak institutions create great incentives to go towards the informal economy, which then starts the noted destructive cycle. A strong rule of law and proper integration of property rights improves the situation of a country regarding its shadow economy (Schneider & Buehn, 2017). It is not difficult to see why this happens and, once again, mirrors our discussion of taxes and regulations. The stronger the system of laws, the less likely people will be to risk engaging in shadow economic activity.

Also, an economy plagued with inefficiencies caused by poor institutions is more likely to see businesses move towards the shadow economy, as operating "above ground" is not efficient (Schneider & Buehn, 2017). To support this view, Canh et al. (2021) find, after controlling for a number of determinants of the shadow economy, that estimates from the WDI for corruption, government effectiveness, political stability and absence of violence / terrorism, regulatory quality, rule of law, and voice and accountability all have significantly negative effects on the size of the shadow economy. Interestingly enough, the effect of corruption on the shadow economy is quite mixed. It is determined by the wealth of the country under study (Dreher & Schneider, 2010). They showed that low-income countries have corruption and their shadow economic apparatus as compliments while they are not in high income countries. On the other hand, Dreher et al. (2009) find that the shadow economy and corruption are, in general, substitutes.

We can extend some of the prior discussion regarding institutions and incentives to the more general idea of deterrence. Deterrence theory has been playing an important role in shadow economy research. Although results vary between negatively impacting or having no impact in the empirical research that does exist, theory suggests that the impact should be highly negative. This view is supported by Feld and Schneider (2010) who find that the greater the probability of detection, the less likely individuals are to work in the underground economy. Given the large amount of incentives individuals have to enter the shadows, controlling this with dissuasion is important. It is also important to note that tax morale plays a role in the shadow activity. This is quite simple in that the greater the willingness to pay taxes based on morality and a positive view of one's government, the less tax evasion / avoidance there will be. An important thing to note is that these variables (namely, tax morale, deterrence, and institutional quality) are interrelated and dependent on each other Feld and Schneider (2010).

A few more determinants of the shadow economy relate to economic development (GDP per capita or GDP growth) Feld and Schneider (2010), the employment composition of an economy (unemployment rate and self-employment rate) (Feld & Schneider, 2010), and the size of the agricultural sector (Hassan & Schneider, 2016). In the case of the first, operating in the shadows is incentivized by a worse economic outlook, causing individuals and businesses to seek out ways to "cost cut" while the economy is in a dip. Regarding employment, a higher rate of unemployment might (1) cause individuals to seek jobs in the shadow economy (this also relates to economic downturns) and (2) a higher unemployment rate is also likely an indicator of the a greater shadow economy. The second reason given also motivates the usage of employment variables as causes and indicators of shadow economy estimates in the MIMIC model (Medina & Schneider, 2017) (discussed in Section 3). Finally, the agricultural sector provides a unique opportunity of underground activity as regulatory environments are weaker in those areas (Hassan & Schneider, 2016).

2.3 Trade and the Shadow Economy

With the background above, it comes time to discuss the possible significance of trade and the size of the underground economy. The upcoming discussion will provide some theoretical reasons for the existence of this relation while also laying the foundations for the choice of variables that will be used. Although trade is often times a subset of the more deeply studied determinants above, it is important to understand whether or not trade can have different or outsized effects. This is motivated by the complexity of international economics, its ability to have large amounts of "spillage effects", and the differing opportunities to act in the shadows. Since trade has a wide variety of procedures, nuances, and potential avenues for informality that are different from those of the local economy, this should be studied separately.

We start by defining what is meant by trade as this will motivate the choice of variables used. Trade will be defined simply along 3 dimensions: (1) trade volume (e.g. imports or exports), (2) trade barriers (e.g. taxes and regulations), and (3) the goods composition (e.g. quality).

To begin our discussion, it is quite well-known that an increase in exports will, in the short term, increase jobs in exporting industries. Not only this, but comparative advantage theory states that, with increased comparative advantage in exports comes greater benefits to trade and thus quality of life (Sirgy et al., 2007). An improvement in the quality of life will lead to a reduced necessity of consumers to engage with the shadow economy. The factors that drive this are multiple. It would not be difficult to see why an increase in a country's exports (in the short term) lead to increases in jobs (thus a reduced incentive for job seekers to enter the informal economy) and sales revenues, which lead to increased tax revenues for the government, and thus to increases in public spending. A number of the aforementioned elements are direct determinants of the size of the shadow economy in the general literature, and these constant cross-effects (i.e. each determinant leading to another determinant) can potentially build towards a reduced shadow economy. An interesting find, however, by Kar and Marjit (2001) is that greater trade openness leads to an increase in informal employment.

Similarly, we see a similar, but possibly more relevant effect with import numbers. Interestingly enough, imports have been studied as less of a factor, yet theoretically they seem to be more important. This relevance comes from multiple dimensions. First of all, we can segment the many benefits of importing (as discussed by Sirgy et al. (2007)) and describe the first category as those that increase tax revenues. These are simply increases in jobs, increases in the sales of the importing sector, and transfers of technology which lead to increases in tax revenues, greater public services, and thus a reduced shadow economy. The other dimension of this analysis is that more imports lead to lower prices and greater access to high quality goods. Clearly lower prices in the formal economy will draw individuals there instead of the shadow economy. This is a simple demand theory of two "businesses", the formal and informal businesses. The preceding analysis indicates that imports may be an important factor affecting shadow economic activity, even beyond the main reasons cited for larger shadow economies.

While there are clear reasons for exports and imports reducing the size of the shadow economy, there are also quite substantial reasons why it might lead to increases. To begin with exports, Sirgy et al. (2007) frame long-term export increases as conducive to national resentment of the exporting country by the importing one, this leads to a chain of events that, in the end, reduces many of the indicators discussed of the size of the shadow economy.

Essentially an increase in exports, will later lead to a decrease as pushback for weakening domestic production. With the similarity to the short-run effects in result but contradiction in the direction, there is now ambiguity as to which direction increased exports should lead the shadow economy. Similarly, increases in imports can undo some of the effects that should have been present. The main avenue for this is in increased competition. Increased competition leads to greater unemployment and lower sales, which in turn reduces tax revenues and starts our cycle increasing shadow economic size again. Not only is this the case, but Tanzi (2002) and Schneider and Enste (2000) discuss competition as a driving force behind businesses moving to the shadow economy. In an attempt to compete with superior importers, they may try to reduce costs by hiring underground workers. They can also purchase more goods from the shadow market. We also have the small addition of decreased employment in importing sectors.

Another volumetric indicator of trade is foreign direct investment. A main through-line effect of FDI is the pressure that it applies on the government to adopt good governance practices. This is done via multinational corporations having a desire to operate in a country, but generally being risk averse and also having flexibility to leave (Blanton et al., 2018). We see this phenomenon as also having a spillage effect, where the practices of large corporations can be learned by smaller ones. This may also include practices that go against using the underground economy or alternatives to it. It is likely that increases in FDI will lead to a reduced shadow economy. Although, this may have a similar, alternative effect as other variables. Namely, that an increase in FDI may indicate greater competition and thus a greater incentive to operate in the shadow economy. A number of studies have tested the relationship between a general index of trade volume and the shadow economy. The following studies, using Dreher’s 2006 index of globalization (Dreher, 2006) (subset into 3 parts: (1) economic, (2) political, and (3) social), have found varying results. Focusing on economic globalization exclusively, a general negative trend is seen, but some model specifications deviate and significance levels are also not homogeneous.

Dreher’s index of economic globalization can be broken down in the following way each component given a non-uniform weight (Figure 2)...

Figure 2: Dreher (2006) Index of Economic Globalization

(i) Flows (all as a % of GDP)	(ii) Restrictions
Trade	Hidden Import Barriers
FDI	Mean Tariff Rate
Portfolio Investment	Taxes on International Trade (as a % of GDP)
Income Payments to Foreign Nationals	Capital Account Restrictions

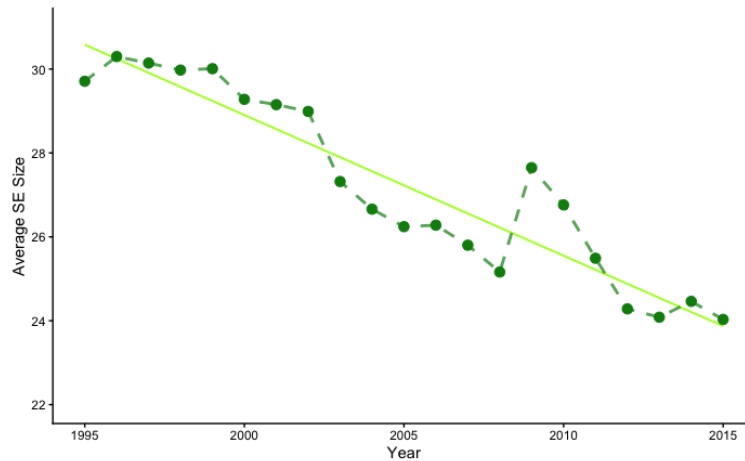
Note that *Income Payments to Foreign Nationals* acts as a proxy for the employment of foreign labor and capital.

Various studies have tested the effects of this index on the shadow economy. To start, Blanton et al. (2018) found that the flows portion of Dreher’s index is negatively related to the shadow economy (to the 10%) when controlling for a number of factors not found to be statistically significant. They did find that the restrictions portion was more significant, yet with similar sized coefficients. Berdiev and Saunoris (2018) simply used the composite index regressed with Dreher’s other components of globalization (political and social) and found that economic globalization had a negative, but statistically insignificant effect. Both of the studies used standard fixed effects estimation and did not include discussions of endogeneity. Fiess and Fugazza (2012) found that under fixed effects estimation their results lined up with those of the two former, yet when using dynamic GMM estimation, that both

the flows and restrictions portions were positively related to the shadow economy, casting doubt on the general usage of fixed effects estimation.

Trade barriers should likely have similar effects to those of other determinants (mainly tax- and regulation-related factors). The imposition of things like trade sanctions could cause a sanctioned country to have alterations in their tax morale as well deterrence. An interesting question relates to which direction this is likely to flow and whether or not the severity of sanctions may play a role. One is likely to argue that if sanctions are imposed that the people of a country may dislike their government for perpetrating the events that caused this to happen in the first place. Rather than this, a country (generally with a decent level of control over their media) can use sanctions, portraying them as targeted against the nation, to spur up national pride and favoritism towards the government (van Bergeijk & Biersteker, 2015). An interesting aspect of sanctions is that total blocks of imports/exports from/to a country are becoming less frequent (Felbermayr et al., 2020). This is likely due to increased dependency in foreign nations for production (either supply chains or specialization). This not only sets the stage for sanctions as an important determinant, but also represent a global trend that may reduce the size of the shadow economy over time.

Figure 3: Average Size of the Shadow Economy By Year



To continue along similar lines, tariff rates and trade taxes serve as widely used barriers to trade. Based on the general theory of taxes and the shadow economy, it is likely that these will have positive effects with the size of the shadow economy. Yet, trade lends additional nuance. To start, an increase in import tariffs may allow local businesses some advantages in that they can offer lower prices all else equal. This reduces competition and thus incentives to operate in the SE. This runs contrary to variables like income tax rates, where Cebula (1997) finds that refraining from increasing the top marginal tax rate can stop the shadow economy from increasing. This type of competition-dampening effect is generally not present with other types of taxes as the only way of taxing foreign competition is through trade taxes.

Another important aspect of the comparison between trade variables and non-trade variables is the reaction to the price sensitivity of their goods. As mentioned earlier, the purchase of informal goods is highly determined by the price of the goods (Williams & Martinez-Perez, 2014). This was named the most important factor when individuals considered whether or not to purchase informal goods. Informal goods can generally be defined as barter of legal goods and services or cash purchases that then go unreported to the gov-

ernment. This can generally reduce the price of goods by eliminating sales tax and reducing income taxes for businesses (therefore allowing them to reduce prices). This can be used as a competitive edge for businesses as well as a way for individuals to alleviate tax burdens placed in other areas. This point of comparison is important as one may argue that an increase in the price of goods for individuals and businesses domestically will have a larger affect than those of imported goods as local businesses (ones that often have the easiest path to the underground economy as they are subject often times to less strict regulations) will have a greater reliance on domestic goods.

A simple hypothesis regarding the quality and diversity of available goods (namely imported goods) is that higher quality goods and a greater diversity of goods will lead to reduced shadow economic activity. It is dubious as to whether or not this will hold. Consequently, we will also give a treatment of this relationship. To focus on purchases, we can see that Williams and Martinez-Perez (2014), in a large survey of European purchasers of informal goods, found that most people had price in mind when purchasing goods in the shadow economy (this focuses on the purchase of legal goods, but without official reporting channels). A large number of purchasers cited inadequacy of goods in the formal sector as a reason for buying without reporting. The numbers are 64% and 38%, respectively. Thus, we can conclude that an improvement in the quality and price of goods imported (often-times coming directly from more imports) are likely to dampen the incentives for purchasing underground. Regarding quality and diversification, Canh and Thanh (2020) find that the diversity (higher index, lower diversification) and quality (higher index, higher quality) of exports both have a concave parabolic relationship with the size of the shadow economy when regressed separately. This suggests the idea that after a certain quality threshold has been surpassed, export quality has negative effects and diversity have positive effects. This could be due to firms which produce higher quality goods are oftentimes multinational, and thus are less likely to operate in local shadow economies. The authors describe this as an increase in market competitiveness leading to a reduced incentive to operate underground. The focus on imports is mainly due to the incentives of buyers as to whether or not they purchase goods in the shadows.

The preceding discussion reveals itself in the mixed results of similar studies thus far. A number of studies have varying levels of significance while all controlling for a different set of variables. To begin, Berdiev and Saunoris (2018) used Dreher’s 2006 and the Dreher et al. (2008) dataset of globalization (broken down into economic, political, and social). When using time and country fixed effects, the researchers find a negative, but statistically insignificant effect of economic globalization on two different estimates of the shadow economy.

3 Data and Variables

The quality of data used is clearly of utmost importance. The following utilizes the most reliable estimates available for the dependent variable and collects independent variables and controls from reputable sources such as the International Monetary Fund, the World Development Indicators data from the World Bank, the Heritage Foundation, and World Integrated Trade Solutions. Some of the variables used are scaled (e.g. per capita metrics) by using WDI’s datasets where needed.

3.1 Dependent Variable

The dataset used for the dependent variables (the size of the shadow economy as a percentage of GDP) is the Multiple Indicators Multiple Causes (MIMIC) model estimations done by Medina and Schneider (2018). These estimates calculated by Schneider and Medina are among the most reliable as the MIMIC model captures the effects of a variety of variables related to the shadow economy versus just a single one (Medina & Schneider, 2018). The MIMIC model includes a two-sided estimation where, on the one hand, a structural equation model is used with variables *Trade Openness*, *GDP Per Capita*, *Unemployment Rate*, *Size of Government*, and *Rule of Law*. On the other hand, a factor-analysis is used with indicators *Currency*, *Labor Force Participation*, and *GDP Per Capita Growth* (Medina & Schneider, 2019). This process yields (via substitutions) a reduced form equation that expresses how the causes and indicators relate (Medina & Schneider, 2018):

$$\mathbf{y}_t = \mathbf{\Pi}\mathbf{x}_t + \mathbf{z}_t \quad (1)$$

This process yields a covariance matrix of observed variables that is a function of the population parameters. This formula then "...minimizes the distance between an observed (sample) covariance matrix and the covariance matrix predicted by the model..." (Schneider & Buehn, 2017). This discussion is important as it motivates the exclusion of certain control variables, namely the ones listed above. This is done to avoid mechanical collinearity. We can see this mechanical collinearity as a two sided equation where certain terms have perfect multicollinearity. We have $\mathbf{y}_t$ as the indicators and $\mathbf{x}_t$ as the causes, which then estimate the size of the shadow economy as discussed above. If these variables were to appear in the equations used to describe this research question, there would be perfect collinearity on both sides of the equation as such

$$se_size(\mathbf{x}_{it}, \mathbf{y}_{it}) = \sum_{x' \in X} \beta_i x'_{it} + \sum_{z \in Z} \beta_i z_{it}$$

For X are variables that appear either in $\mathbf{y}$ or $\mathbf{x}$, and for Z are variables that do not appear in $\mathbf{y}$ nor $\mathbf{x}$. To avoid this, variables that appear in Medina and Schneider (2018) (i.e. $\mathbf{y}$ nor $\mathbf{x}$) are not used in the equations of this study.

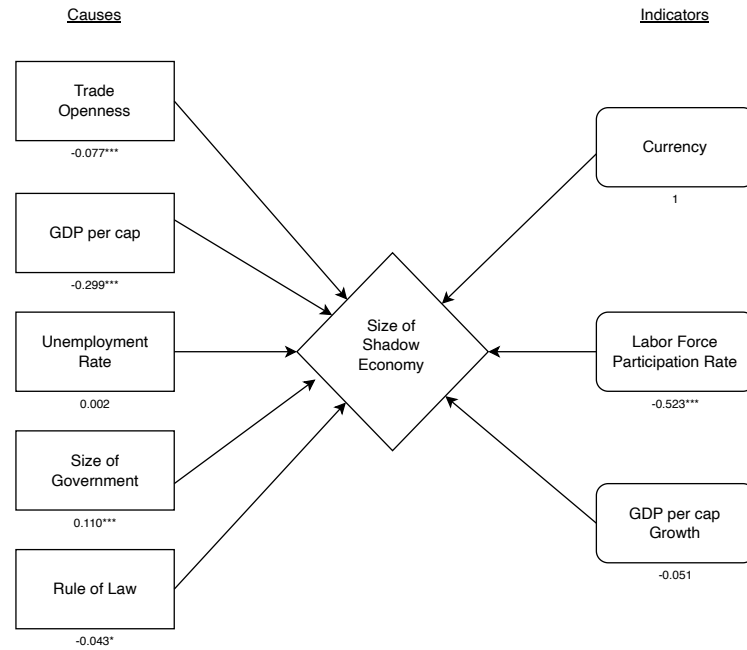
The usage of *Trade Openness* as an observed causal variable aids in our discussion by its significance in the model estimated by Medina and Schneider (2018). The coefficient of *Trade Openness* is statistically significant (ranges between 5% and 1%) and is negative on all model specifications (ranges from -0.097 and -0.062). Thus, for every increase in the trade openness index (defined as (ImportsExports)/GDP), a nation can expect to see around a 0.08 point decrease in the size of their shadow economy. Figure 4 shows an example of the hypothesized path for the MIMIC estimator as well as the causes and indicators accompanying it.

There are a few limitations of the model such as a difficulty of excluding traditional crime, unstable coefficients when the model specifics or sample size are changed, and benchmarking procedures are not very robust when obtaining "real world figures" (Medina & Schneider, 2018).

3.2 Independent Variables

Maintaining along the lines of the 3 pillars of trade defined at the start of section 2.3, we use the following variables.

Figure 4: MIMIC Model Hypothesized Path



Recreated from Medina and Schneider (2019)

It is simple enough to address the first pillar. Measures of trading volume are quite well-captured by indices like **Trade Openness**. The section above outlines some results regarding that taken from Medina and Schneider (2019).

A metric that will be used here is the amount of foreign direct investment in the country (**FDI**). This is used to see if the volume of money entering an economy for investment by individuals abroad is related to the shadow economy. This measure is taken as percentage of GDP and is among the more instrumental variables in analyzing international economics.

Onto taxes, the weighted average applied tariff rate of the countries (called **Applied Tariff Rate** in the model) is used to capture the premier tax rate indicator of trade. This variable, as the name suggests, is the weighted average (all products) of applied import tariffs levied by each country to their partner countries. This data was taken from the World Development Indicator's (WDI) "Tariff rate, applied, weighted mean, all products (%)" dataset. This variable is important because, as discussed earlier, taxation is a major driver of shadow economic activity. The attempt here is to understand whether or not tax-trade variables have an outsized effect on the shadow economy.

Taxes on international trade, called **Trade Taxes** in the model, encompass all taxes levied on trade. The Australian Bureau of Statistics defines taxes on international trade as "taxes that become payable when goods cross the national or customs frontiers of the economic territory..." (2015). They further fragment this into customs duties on imports and exports, agricultural produce export taxes, and others. This variable is similar to the other one, but captures the relational monetary value collected from taxes, not what effect the current rates of taxation have on a countries shadow economic movements. This variable is defined as trade taxes as a percentage of GDP and is used to capture the significance of the trade barriers put in place by these taxes. This data was taken from the IMF's World Revenue Longitudinal Database.

An additional, but more implicit barrier to trade, is the price of goods imported. This is captured via the **Import Unit Value Index** (or Import UVI) (WDI). This is calculated as the weighted average relative price ratios and is fixed at 2015 prices. This allows one to capture increases in relative prices to capture the effects of individuals moving to the shadow economy as prices go up. It may be useful to see if price changes for goods imported will be different from domestic price changes (see Control Variables for the comparison). This can be the case as increased usage of international supply change, may allow non-domestic producers more flexibility. Local business are more likely to make purchases domestically, thus lending them less protection from price changes. There also may be the case of differences in preferences regarding supplementing goods. Since domestically produced goods are often-times more of a necessity (especially in less developed nations), people's tendencies to supplement may change depending on whether or not local or international goods change prices.

To conclude the barriers to trade, the presence of **Trade Sanctions** comes included as a shifter. This is captured with an import sanction imposed, an export sanction imposed, and a combination of both (note that this does not include whether or not the sanctions are partial or total. It is import to delineate between import versus export as import tariffs can represent a necessity to supplement purchases (if say an imported good were optimal, then an individual may look towards obtaining the good in the shadow economy to offset either the price or the quality change that is now necessary. Both sanctions have bases regarding competition. For the case of imports, if a country were have their imports blocked, competition among the domestic firms would be reduced as international producers no longer have access to that specific market. For exports, we see a reduction in profits for local domestic firms, leading them to become less competitive. These mirror our through-line discussion of the incentives for businesses to utilize the underground economy. The interaction between these two (representing "both") is simply useful in seeing which of these effects take over or if they move in unison. The data for this variable was taken from Felbermayr et al.'s Global Sanctions Data Base (GSDB). For the purposes of simplicity, only dummy variables of whether or not a country had an import sanction, export sanction, or both imposed on it for a specific year. No inclusion of the success rate or whether or not a partial or total sanction was enacted.

And finally, we want to capture the quality of goods that the individuals in a country have access to pre-trade by taking the weighted average of their reveal comparative advantages across a wide variety of product classes (called **Weighted Average RCA**). The reveal comparative advantage (RCA) of a specific product can be defined by the formula...

$$RCA_{Ai} = \frac{\frac{X_{Ai}}{\sum_{j \in P} X_{Aj}}}{\frac{X_{wi}}{\sum_{j \in P} X_{wj}}}$$

Where...

- P is the set of all products (with $i \in P$),
- X_{Ai} is the country A's exports of product i,
- X_{wi} is the world's exports of product i,
- $\sum_{j \in P} X_{Aj}$ is the country A's total exports (of all products j in P), and
- $\sum_{j \in P} X_{wj}$ is the world's total exports (of all products j in P).

This formula simply represents the ratio of exports of a product versus total exports of a country relative to the same ratio in the world. The rational behind this variable is that if a country has an advantage in producing a variety of products, then individuals will be less

willing to supplement purchases in the shadow economy. Both the RCA's of each product category and exports of the same product categories were taken from the World Integrated Trade Solutions database. It is important to note that the product classes chosen will have some overlap (this was seen when summing them together lead to values greater than export numbers) which can lead to imperfect weighted averages.

3.3 Control Variables

The main justification for the control variables is the discussions done by Schneider and Enste (2000) and Schneider and Buehn (2017). These are summarized in Figure 5.

Variable	Description
Sales Tax, Social Contributions, Property Tax, and Income Tax	The inserting of these tax variables into the regression analysis is done not only to control for their effects on the shadow economy, but to stand as a comparator for trade taxes. Different types of tax categories can have different effects on shadow economies depending on the degree / direction that they influence things like consumer purchases, employment, and the over burden of the tax. A detailed discussion of this in general is done in the literature review. Source: IMF - World Revenue Longitudinal Database
Inflation	Inflation is included as a control variable as price changes may induce individuals to go to the underground economy. Higher prices in the economy may lead to individuals to seek out additional work (this is often done in the shadows), it may also cause them to supplement goods purchases, or even to avoid paying certain taxes. This is also used as a basis of comparison with the Import UVI variable. The deflator was chosen over CPI because the CPI doesn't include goods that businesses or corporations may purchase, so to include these the overall value of the currency is chosen. The values were taken from the a dataset which had different base years for each country and each country was shifted to have a base year of 1995 as that is the first year of the available data for the dependent variable. Source: WDI - GDP Deflator (base year varies by country)
Agricultural Size	Based on the results of Hassan and Schneider (2016) a larger agricultural sector is known to increase informality as regulation (by local governments) is difficult in this sector. Source: WDI - Agriculture, Forestry, and Fishing, value added (% of GDP)

Figure 5: Control Variables

4 Results

4.1 Simple Linear Models

To begin, we used the following simple linear model as a starting point for the analysis,

$$se_size_{it} = \sum_{x \in X} \beta_i x_{it} + \sum_{y \in Y} \gamma_i y_{it} + u \quad (2)$$

for X is the set of independent variables and Y is the set of control variables as defined in the methodology. We report the results of the initial-stage, simple models below (Table 1 and Table 2). This begins with pooled OLS, random effects, and country-fixed-effects. The following doesn't include nonlinearities or interaction terms. This is done to mirror the current research on the topic to see if the following are more robust. To start, we can use a

Hausman test to determine which model might present better results.

Note: a single observation was removed as it had an Applied Tariff Rate of 400 which has altered some of the linear and nonlinear estimates.

data: form_with_controls_w_time

chisq = 1718.8, df = 14, p-value ; 0.00000000000000022

alternative hypothesis: one model is inconsistent

From this we deduce that the random effects model is will not have consistent estimators. The headings in the table outline which model is used (Table 1) (note that the intercepts for the fixed effects models were recovered using a weighted mean of each country's intercept). Figure 6 presents the correlation matrix and Figure 7 presents the variance inflation factors for model (*Pooled*) with control variables. Also results regarding each individual independent variable's relation to the shadow economy (time-fixed effects) are reported in Table 2.

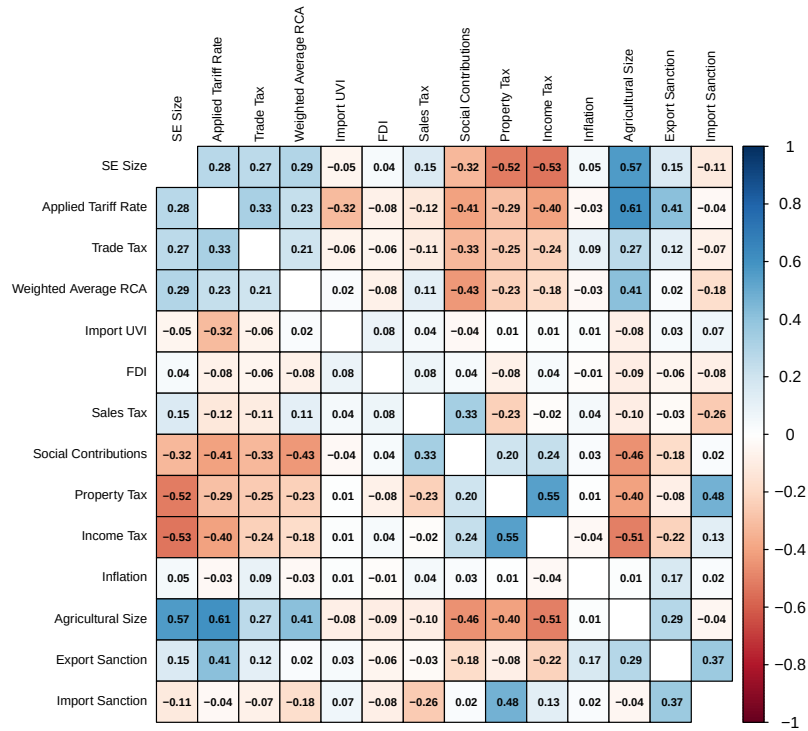


Figure 6: Correlation Matrix

Applied Tariff Rate	Trade Tax	Weighted Average RCA	Import UVI Growth Rate
1.6683	1.2441	1.5122	1.0370
Sales Tax	Social Contributions	Income Tax	Property Tax
1.2906	1.803978	1.6262	1.5356
	CPI	Agirculture Size	
	1.2121	2.0852	

Figure 7: Variance Inflation Factors

Table 1: Simple Linear Model Results

	<i>Dependent variable:</i>					
	SE Size					
	<i>OLS</i>		<i>panel linear</i>			
	(Pooled)	(Pooled)	(Random)	(Random)	(Fixed)	(Fixed)
(Intercept)	20.725*** (1.990)	31.214*** (2.169)	32.001*** (3.199)	36.225*** (6.914)	31.368*** (0.711)	36.61689*** (1.527)
Applied Tariff Rate	0.560*** (0.145)	-0.446*** (0.142)	0.301*** (0.067)	0.170** (0.076)	0.296*** (0.065)	0.175*** (0.065)
Trade Tax	0.957*** (0.134)	0.456*** (0.081)	-0.138 (0.094)	-0.094 (0.084)	-0.169* (0.096)	-0.117 (0.075)
Import UVI	-0.042** (0.018)	-0.058*** (0.014)	-0.062*** (0.011)	-0.047*** (0.010)	-0.062*** (0.011)	-0.045*** (0.009)
Weighted Average RCA	2.147*** (0.421)	-0.166 (0.381)	-0.652 (0.853)	-0.576 (0.874)	-0.754 (0.879)	-0.589 (0.820)
FDI	0.033*** (0.007)	0.028*** (0.006)	0.007 (0.007)	0.009 (0.008)	0.007 (0.007)	0.008 (0.008)
Import Sanction	-3.710** (1.672)	3.519*** (1.269)	-0.661 (0.624)	-0.852* (0.514)	-0.635 (0.641)	-0.717 (0.487)
Export Sanction	-0.754 (1.965)	-1.339 (1.904)	-0.704 (0.695)	-1.161 (0.768)	-0.728 (0.695)	-1.279 (0.807)
Import Sanction $\times$ Export Sanction	11.817*** (3.018)	-2.020 (2.621)	2.122 (1.321)	3.116*** (1.010)	2.086 (1.326)	3.175*** (1.032)
Sales Tax		0.797*** (0.157)		-0.368* (0.194)		-0.445*** (0.169)
Social Contributions		-0.353*** (0.081)		-0.335 (0.448)		-0.219 (0.580)
Property Tax		-2.795*** (0.361)		0.489 (0.570)		0.855 (0.537)
Income Tax		-0.573*** (0.064)		-0.415*** (0.159)		-0.355** (0.174)
Inflation		0.00002*** (0.00001)		0.00001 (0.00002)		-0.003** (0.001)
Agricultural Size		0.803*** (0.090)		0.254** (0.105)		0.121 (0.133)
Observations	942	856	942	856	942	856
R ²	0.191	0.502	0.369	0.395	0.378	0.424
Adjusted R ²	0.184	0.494	0.364	0.385	0.323	0.365
Fixed Effects	No	No	No	No	Yes	Yes
F Statistic	27.471***	60.547***	523.168***	500.036***	65.654***	40.722***

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 2: Simple Linear Model Results for each Independent Variable

	<i>Dependent variable:</i>								
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Applied Tariff Rate	0.327*** (0.074)								0.100** (0.050)
Trad Tax		0.125 (0.219)							−0.115* (0.063)
Import UVI			−0.055*** (0.009)						−0.010 (0.023)
Weighted Average RCA				−0.445 (0.933)					−0.312 (0.703)
FDI					0.008 (0.008)				0.007 (0.008)
Import Sanction						0.443 (0.735)			−0.030 (0.572)
Export Sanction							1.183 (0.860)		−0.408 (0.777)
Import Sanction × Export Sanction								2.302** (0.959)	1.823* (1.043)
Sales Tax	−0.327 (0.232)	−0.582** (0.281)	−0.553** (0.240)	−0.564** (0.234)	−0.637** (0.268)	−0.600** (0.279)	−0.600** (0.277)	−0.603** (0.279)	−0.373*** (0.132)
Social Contributions	−0.342 (0.606)	−0.431 (0.606)	−0.199 (0.615)	−0.363 (0.570)	−0.443 (0.611)	−0.413 (0.611)	−0.384 (0.620)	−0.412 (0.610)	−0.126 (0.597)
Property Tax	0.458 (0.464)	0.206 (0.502)	0.753 (0.631)	0.405 (0.482)	0.250 (0.537)	0.181 (0.493)	0.084 (0.468)	−0.002 (0.439)	1.129** (0.553)
Income Tax	−0.071 (0.195)	−0.232 (0.222)	−0.415** (0.193)	−0.247 (0.220)	−0.246 (0.224)	−0.233 (0.222)	−0.237 (0.220)	−0.250 (0.216)	−0.240 (0.188)
Inflation	−0.001** (0.0004)	−0.001* (0.0004)	−0.0005 (0.0003)	−0.004** (0.002)	−0.001 (0.0004)	−0.001* (0.0004)	−0.001* (0.0004)	−0.001* (0.0004)	−0.002** (0.001)
Agricultural Size	0.358*** (0.128)	0.477*** (0.129)	0.171 (0.121)	0.453*** (0.131)	0.482*** (0.130)	0.494*** (0.131)	0.491*** (0.130)	0.491*** (0.127)	0.051 (0.129)
Observations	914	911	868	903	914	914	914	914	856
R ²	0.269	0.215	0.359	0.271	0.218	0.215	0.221	0.232	0.492
Adjusted R ²	0.205	0.147	0.300	0.208	0.151	0.147	0.153	0.166	0.425
F Statistic	44.052***	32.788***	63.428***	44.039***	33.547***	32.928***	34.046***	36.327***	21.514***

Note:

*p<0.1; **p<0.05; ***p<0.01

To confirm that the model does not include groups of variables lacking explanatory power, joint hypothesis tests were conducted. A joint hypothesis test for all of the tax variables that are non-significant shows that they are group-wise statistically significant. What can be seen is that most of the tax variables (aside from *Property Taxes*) are negatively related to the shadow economy (note that we are referring to the results of the fixed effects models). This goes against prevalent theory of taxation and the shadow economy, but misses out on certain factors. These values are calculated as tax revenues as a percentage of GDP, not the tax rates. The reason why these signs might be negative is these values may be implicit indicators of countries' effectiveness in collecting that type of tax. The quality of regulations and tax-collecting ability has been found to be negatively related to the shadow economy (Schneider & Enste, 2000). The reason why property taxes may show the expected relationship is that the collection of property taxes is generally more straight-forward and is less subject to avoidance / evasion. In this case, tax volumes collected is more likely to be to reflect the tax rates / payments per individual.

We also test for some of the related trade / pricing variables (Weighted Average Revealed Comparative Advantage, Import UVI Growth Rate, and the CPI) and find that they are not important determinants of the shadow economy, yet it seems highly plausible that 2 measures of consumer prices (that may not have significant overlap) are important determinants of the shadow economy. The results are reported in Figure 8.

Figure 8: Tax Variables and Trade/Pricing Joint Significance Tests

Linear hypothesis test: Trade Tax = Property Tax = Social Contributions = Income Tax = 0

Model 1: restricted model

Model 2: full model

	Res.Df	Df	Chisq	Pr(>Chisq)
1	848			
2	844	4	21.124	0.0002993***

Linear hypothesis test: Import UVI = Inflation = Weighted Average RCA = 0

Model 1: restricted model

Model 2: full model

	Res.Df	Df	Chisq	Pr(>Chisq)
1	848			
2	844	4	21.124	0.0002993***

Linear hypothesis test: Import UVI = Inflation = Weighted Average RCA = 0

Model 1: restricted model

Model 2: full model

	Res.Df	Df	Chisq	Pr(>Chisq)
1	719			
2	716	3	4.0227	0.259

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

4.2 Nonlinearities

An important aspect lacking in other research regarding trade and the shadow economy is the usage of nonlinearities. A preliminary look at the graphs of the taxation volumes and the shadow economy sizes shows that there may be significant nonlinearities (Figure 9).

Theoretically, taxes rate are known to have positive effects on the shadow economy (thus by extension tax volumes in many instances, but there may be an effect where this direction switches at some point. This "turn around" can be brought about due to efficiency gains with increased tax volumes. Taxation is likely to incese the size of shadow economies as it gives individuals incentives to avoid / evade taxes, yet after a certain point, the extra revenues given to the government from that specific tax source may lead to improvements in government efficacy with regards to preventing the shadow economy. Additional tax revenues also lead to higher institutional quality which is known to reduce the size of the shadow economy (Schneider & Enste, 2000).

The reverse direction shift may also exist, where a level of taxation has a negative effect until a parabolic vertex. This may happen simply as a pull between tax morale and encentives. At low volumes of taxation, people may be more willing to pitch in extra as a service to their nation, but after a certain point, this positive morale may be overridden by the benefits of not paying taxes. The inclusion of these nonlinearities is to capture these effects. An additional, but also less prevelent reason, is simply to capture potential diminishing returns to taxation as a way to reduce the size of shadow economies. The inclusion of nonlinearities is done to all tax-related variables for trade (i.e, **Applied Tariff Rate** and **Trade Taxes**). The results are reported in Table 9.

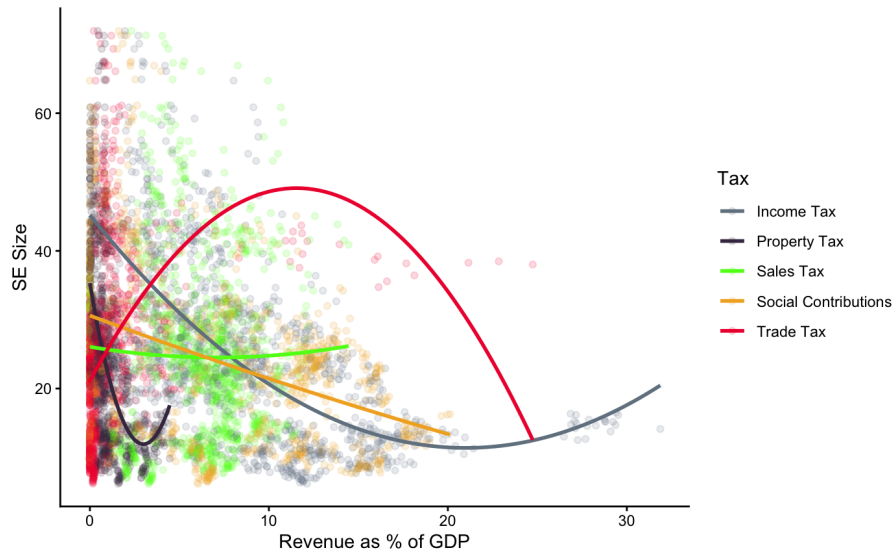


Figure 9: Visualization of Tax Volume Nonlinearities

Despite a potentially strong theoretical backing, the results do not with much certainty reveal significant nonlinearities. The R^2 and $\bar{R}^2$ are not very different with the inclusion of the nonlinearities and their coefficients are statistically zero. Unsurprisingly, the significance of these variables are lost as the inclusion of these terms exacerbates mutlicollinearity problems. From this one can deduce that nonlinearities do not play a significant role in trade-related taxation's explanatory power.

4.3 Interactions

Another model specification lacking in the research on trade and the shadow economy is the usage of interactions. Since most of the papers on this topic have used Dreher's (2006) index, interacting with specific aspects of trade are not possible. The results are in Table 4.

Table 3: Nonlinearities Included

	<i>Dependent variable:</i>			
	se_size			
	(1)	(2)	(3)	(4)
applied_tariff_rate	0.100** (0.050)	0.126 (0.132)	0.102** (0.046)	0.131 (0.125)
I(applied_tariff_rate^2)		−0.001 (0.005)		−0.001 (0.004)
trade_tax	−0.115* (0.063)	−0.120* (0.063)	−0.147 (0.251)	−0.162 (0.241)
I(trade_tax^2)			0.001 (0.008)	0.002 (0.007)
import_uvi	−0.010 (0.023)	−0.011 (0.024)	−0.010 (0.023)	−0.011 (0.024)
wa_rca	−0.312 (0.703)	−0.312 (0.701)	−0.308 (0.699)	−0.306 (0.697)
fdi	0.007 (0.008)	0.007 (0.008)	0.007 (0.008)	0.007 (0.008)
import_sanction	−0.030 (0.572)	−0.032 (0.568)	−0.027 (0.574)	−0.028 (0.569)
Export Sanction	−0.408 (0.777)	−0.404 (0.775)	−0.395 (0.795)	−0.386 (0.790)
I(import_sanction × export_sanction)	1.823* (1.043)	1.850* (1.065)	1.800* (1.066)	1.822* (1.084)
sales_tax	−0.373*** (0.132)	−0.368*** (0.125)	−0.371*** (0.133)	−0.366*** (0.126)
social_contributions	−0.126 (0.597)	−0.127 (0.598)	−0.124 (0.598)	−0.125 (0.598)
property_tax	1.129** (0.553)	1.126** (0.552)	1.133** (0.552)	1.130** (0.551)
income_tax	−0.240 (0.188)	−0.238 (0.186)	−0.240 (0.188)	−0.239 (0.187)
inflation	−0.002** (0.001)	−0.002** (0.001)	−0.002** (0.001)	−0.002** (0.001)
size_agri	0.051 (0.129)	0.051 (0.128)	0.052 (0.129)	0.053 (0.127)
Observations	856	856	856	856
R ²	0.492	0.492	0.492	0.492
Adjusted R ²	0.425	0.425	0.424	0.424
F Statistic	21.514***	20.878***	20.873***	20.274***

Note:

*p<0.1; **p<0.05; ***p<0.01

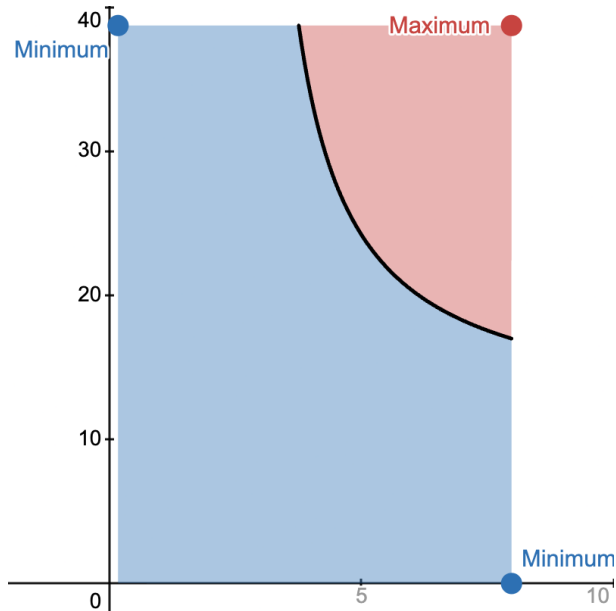
The first interaction under discussion is that of the **Applied Tariff Rate** and **Import Sanctions**. It seems plausible that the effects of import tariffs on the shadow economy will depend on the volume of imports to an extent. If a country is under import sanctions, than it will have less products to import as well as a lower product variety, thus tariffs will have less of an affect on competition and thus a reduced effect on the shadow economy. Under import sanctions, countries will have reduced competition, making the tariff rates a less effective tool to reduce this competition. This is not found to be significant.

An interaction between **Weighted Average Revealed Comparative Advantage (RCA)** and **Agricultural Size** is also explored. A justification for this can run along the lines of the following: if countries have a significant comparative advantage in a sector like agricultural (say their population is larger or they have more agriculture-related factors like land), then it could take advantage of gains from trade. Not only will these gains from trade reduce incentives to engage in the shadows, but the government, with the knowledge of this industries' strength, may attempt to increase deterrance efforts or alter the tax rates in such a way that the government can collect additional revenues, thus allowing them to reduce the size of the underground economy.

The interaction between **Weighted Average RCA** and **Agricultural Size** is not only significant, but has added significance to the comparative advantage variable. The result of this is a decision boundary of sorts, where a nation under a certain threshold of the size of their agricultural sector and their weighted average revealed comparative advantage can have their shadow economy reduced by divesting from their agricultural sectors. This boundary is governed by the equation

$$0 = -1.621x + 0.143xy - 0.380y$$

for $x = \text{Weighted Average RCA}$ and $y = \text{Agricultural Size}$.



Note: blue is negative effects, red is positive effects, the black line is 0, the bounds are the sample mins and maxes.

Figure 10: Weighted Average RCA and Agricultural Size Boundary

We can visually see this decision boundary. Note that the 2 minimums appear at the maximum-minimum pairs of the variables (e.g. minimum of **Weighted Average RCA**

and maximum of **Agricultural Size**). A sector that has a comparative advantage will likely draw more labor to it and thus leave greater opportunities for shadow activity. Since the agricultural sector is notably difficult to regulate with regards to collecting taxation, it could be the case that a greater comparative advantage is drawing increased shadow activity when the size of the agricultural sector is high and thus even more difficult to regulate. Despite this, the benefits of comparative advantage seem to have negative effects in most cases.

5 Additional Model Specifications

A large body of research has suggested the prevalence of the "shadow cycle" presented in the introduction and literature review. This cycle is described as the shadow economy feeding into itself in a viscous cycle, wherein shadow activity removes revenue from governments while also costing them in deterrence efforts. This then requires the government to raise tax rates to make up the difference, thus increasing shadow activity further. The following model specifications address this phenomenon.

5.1 Endogeneity

Endogeneity has needed addressing and is clearly present due to the cycle referenced throughout the paper. To start, however, omitted variable bias is brought into focus. Based on the prevalent literature, the regression variables presented, as well as the variables included in the Medina and Schneider (2018) estimates cover a lot, but not all of the variables that are well-known to relate to the shadow economy. Mainly, these variables are tax rates (excluding **Applied Tariff Rates** clearly), tax morale, deterrence, and others. Thus, one can easily see how tariff rates and tax volumes would suffer from omitted variable bias. Since instrumenting 6 endogenous variables with 5 variables assumed exogenous is not feasible, a two-staged copula correction (2sCOPE) method will be used to control for endogeneity.

The 2sCOPE method for endogeneity correction has recently gained traction in the world of marketing and management and is quickly being introduced as a quality method for dealing with endogeneity. This method has been used by hundreds of papers and over 10% having been used for economics or closely-related disciplines (Qian et al., 2024). As outlined above, the general motivation for usage of copula correction is when sufficiently high-quality IVs are not available. Additionally, copulas can be used in mixed models (where both copulas and IV estimation are used), or as a robustness test for instruments (Qian et al., 2024).

The general approach is to control for the endogenous portions of our assumed endogenous variables by (1) regressing the copulas of each endogenous variables with the copulas of all of our exogenous variables and (2) using the residuals as a control variable. If the coefficient for this variable is zero, the hypothesis that it is not endogenous cannot be rejected (Qian et al., n.d.); (Qian et al., 2024). A main caveat of utilizing 2sCOPE is that it requires the satisfaction of various assumption and boundary conditions. They are summarized below. Another caveat is that it is not robust towards nonlinearities, interactions, and dummy variables so they have been removed from the analysis.

Using the specifications above, these assumptions are tested and empirics support them, thus copula correction is likely to be robust. Below is a set of figures regarding *Boundary Condition 1* as well as *Boundary Condition 2*. We can see that the distribution is symmetric according to the Miao, Gel, and Gastwirth (MGG) test, yet the distribution is nonnormal.

Table 4: Interaction Terms

	<i>Dependent variable:</i>			
	se_size			
	(1)	(2)	(3)	(4)
applied_tariff_rate	0.100** (0.050)	0.139* (0.077)	0.103** (0.045)	0.125* (0.071)
I(applied_tariff_rate *import_sanction)		−0.086 (0.085)		−0.050 (0.082)
trade_tax	−0.115* (0.063)	−0.136* (0.070)	−0.136** (0.060)	−0.148** (0.072)
import_uvi	−0.010 (0.023)	−0.013 (0.023)	−0.017 (0.022)	−0.018 (0.023)
wa_rca	−0.312 (0.703)	−0.312 (0.698)	−1.639* (0.852)	−1.621* (0.845)
I(wa_rca *size_agri)			0.145*** (0.042)	0.143*** (0.042)
fdi	0.007 (0.008)	0.007 (0.008)	0.007 (0.007)	0.007 (0.007)
import_sanction	−0.030 (0.572)	0.196 (0.611)	−0.122 (0.548)	0.010 (0.559)
export_sanction	−0.408 (0.777)	−0.402 (0.770)	−0.257 (0.779)	−0.256 (0.775)
I(import_sanction *export_sanction)	1.823* (1.043)	2.083* (1.113)	1.546 (1.038)	1.701 (1.119)
sales_tax	−0.373*** (0.132)	−0.347*** (0.125)	−0.357*** (0.133)	−0.343*** (0.126)
social_contributions	−0.126 (0.597)	−0.118 (0.595)	−0.085 (0.580)	−0.081 (0.579)
property_tax	1.129** (0.553)	1.086** (0.540)	0.986* (0.553)	0.963* (0.545)
income_tax	−0.240 (0.188)	−0.239 (0.188)	−0.242 (0.190)	−0.241 (0.189)
inflation	−0.002** (0.001)	−0.002** (0.001)	−0.002** (0.001)	−0.002** (0.001)
size_agri	0.051 (0.129)	0.064 (0.125)	−0.393** (0.200)	−0.380* (0.199)
Observations	856	856	856	856
R ²	0.492	0.493	0.513	0.513
Adjusted R ²	0.425	0.426	0.448	0.448
F Statistic	21.514***	20.967***	22.681***	22.055***

Note:

*p<0.1; **p<0.05; ***p<0.01

Figure 11: Assumptions of 2sCOPE that must be satisfied

Assumption 1	Either the error E_i or specifically its endogenous part U_i is normally distributed. This is not strictly necessary as copula correction is robust to symmetric distributions of U . An inspection of residuals and the usage of Boundary Condition 1 can aid in testing for this.
Assumption 2	Either $(P, E) W$ or $(P, U) W$ follows a Gaussian Copula. Similar to the above in addressing it.
Assumption 3	Full rank of all regressors and $Cov(W, E) = 0$.
Assumption 4	P is continuous and the copula term $C_{p w}$ is linearly independent of $(1, P, W)$. This requires the nonnormality of P or at least one exogenous regressors in W . This can be checked with Boundary Condition 2.
Boundary Condition 1	Inspect the inflation of standard errors. If inflation > 6 indicates likely GC dependence model misspecification.
Boundary Condition 2	The continuous P is nonnormal (normality test $p < 0.05$) or one exogenous W is nonnormal (normality test $p < 0.001$) and sufficiently relevant (first-stage F statistic > 10).
Boundary Condition 3	A sample size of 300 is recommended.

Summarized from Qian et al. (2024)

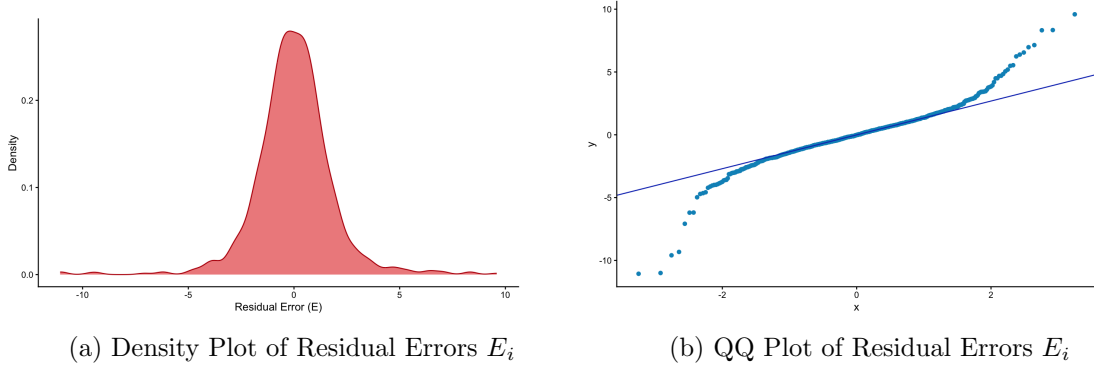


Figure 12: Normality / Symmetry of Errors

m-out-of-n bootstrap symmetry test by Miao, Gel, and Gastwirth (2006) data:
res Test statistic = 0.38337, p-value = 0.766 alternative hypothesis: the distribution is asymmetric.

Regarding the *Boundary Condition 2*, it can be seen that all chosen endogenous variables are nonnormal and thus this has been satisfied. Given the satisfaction of the boundary conditions, the results of the linear model with time-fixed effects with 2sCOPE are presented in Table 5. This application results in changes to the model and thus reveals very prevalent endogeneity within the model, at least with regards to omitted variable bias and measurement error (which is not assumed a major issue here given the data sources used, this could only apply to the (Medina & Schneider, 2018) MIMIC estimates).

From the results we can not only see that the model has changed substantially, but which terms had large endogeneity problems by evaluating the size of the copula term. Theory regarding shadow economies states that **Social Contributions** are among the largest indicators of underground activity, yet they seemed of middling significance in the models discussed previously. Now that the endogenous portion of it has been controlled for, we see its significance on full display. Although, seeing the significance of this variable is important,

Variable	Applied Tariff Rate	Trade	Income	Social	Sales	Property
W =	0.75952	0.38205	0.90795	0.90509	0.98733	0.84709
p-value	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
Normal	No	No	No	No	No	No

Figure 13: Shapiro-Wilk Normality Test for Endogenous Variables

Table 5: 2sCOPE Procedure for Endogeneity

<i>Dependent variable:</i>			
	SE Size		
	(Linear Terms)	(Copula Terms)	
Applied Tariff Rate	−0.550*** (0.200)	Copula	0.501 (0.664)
Trade Tax	0.674*** (0.122)	Copula	−0.635 (0.602)
Import UVI	−0.015 (0.042)		
Weighted Average RCA	−2.646*** (0.574)		
FDI	0.020** (0.008)		
Income Tax	−1.000*** (0.983)	Copula	3.088***
Property Tax	1.080* (0.652)	Copula	−3.600*** (0.790)
Social Contributions	−1.903*** (0.232)	Copula	9.917*** (1.452)
Sales Tax	0.350 (0.587)	Copula	0.511 (1.464)
Inflation	0.000*** (0.000)		
Agricultural Size	0.542*** (0.114)		
Observations	758		
R ²	0.552		
Adjusted R ²	0.532		
F Statistic	27.07***		

Note: *p<0.1; **p<0.05; ***p<0.01

its sign is the opposite of what is expected.

An interesting aspect of this is that the time variables have lost a lot of their explanatory power. In the previous models, nearly all years were statistically significant to the 5% at the very least and had coefficients that reached up to 9. This is no longer the case, suggesting that a lot of the global trends that were captured in the time-fixed effects existed in the endogenous error of the tax variables. These could represent general trends toward better governance as knowledge of how to better govern as well as experience gained from international organizations in helping countries to effectively collect taxes (i.e. improvements in deterrence efforts) or general improvements in tax morale over time.

6 Conclusions and Policy

6.1 Conclusions

An initial conclusion that can be drawn from this analysis is that applied tariff rates have a negative effects on the size of shadow economies. This goes against the prevailing theory that tax rates have positive effects on underground activity. An alternative to this suggests that it is not the tax rates, but the ability to collect these taxes that has a greater influence on the shadow economy and that assertion is supported here. Since the revenues from tariffs are generally harder to avoid, it acts as a reliable source of non-shadow-economy-increasing revenue.

The sign of **Applied Tariff Rates** is likely negative due to the fact that these taxes can aid in deterrence efforts for other tax variables. This suggests that if countries want to focus on reducing informality, they could increase tariffs while lowering other taxes on trade (relationship is positive as seen by the coefficient for **Trade Tax**).

It is likely that other taxes on trade increase informality because they reduce the competitiveness of companies that rely more on exports to make a profit. A major throughline in this paper has been that competitiveness (or lackthereof) is an important incetivisor for informality. Tariff rates and trade tax volumes having opposite signs suggest that nations could reduce underground activity by shifting competitiveness to local firms by increasing import tariffs while lower taxes on trade for domestic producers. The relationship between these trade-related tax variables is further supported by foreign direct investment's positive relationship with shadow economies. Increased investment into domestic firms from abroad poses a threat to local firms that may feel pressure to make up the difference with shadow activity.

A clear conclusion that can be seen across many of the models presented earlier is that the imposition of both import and export sanctions on a country increases the size of the shadow economy (each also individually increase informality, but become negative with the inclusion of the joint interaction). The relation governing the imposition of sanctions and the shadow economy (taken from Table 1, last column) is

$$-0.717 \times \text{Import} - 1.279 \times \text{Export} + 3.175 \times \text{Both}$$

with partial effects...

$$\text{Imports: } -0.717 + 3.175 \times \text{Export}$$

$$\text{Exports: } -1.279 + 3.175 \times \text{Import}$$

$$\text{Both: } 1.179$$

What we see with the above analysis is that the imposition of an import or an export tariff individually will reduce informality, but having both will significantly increase the size of a country's shadow economy. This equation likely stems from the multicollinearity between the variables. Including these individually shows that all variables have a positive relationship (Table 2) and that, with both, the results are quite similar and the type of sanction does not have a major impact, just that they exist. The positivity of these variables can be due to a variety of reasons, among which are: reduced product quality and diversity, depressed tax morale as citizens may negatively view their government for subjecting them to the impacts of the sanction, a reduction in the ability to prevent tax avoidance / evasion as resources need to be diverted towards dealing with the ramifications of the sanctions, and more.

Finally, the interaction done between **Weighted Average RCA** and **Agricultural Size** shows that for most nations, having a large amount of comparative advantage does well to reduce shadow economic activity. This is likely due to general welfare gains from trade and increased tax revenues from trade (at no cost from raising rates). A peculiarity here is that with both variables being relatively high, the overall effects are positive with underground activity. This is likely due to the large amount of labor that may enter this sector if it does well both domestically (by size) and internationally (by comparative advantage). Since the agricultural sector is notoriously difficult to regulate, it is subject to exacerbated amounts of informality (Hassan and Schneider, 2016). Thus with a larger labor supply (workers in agriculture may be more subject to informal wages) and greater difficulty deterring tax evasion / avoidance, comes increases in informality.

6.2 Policy Implications

To place the above into a more practical lense, countries can alter trade variables to reduce the size of their informal sectors by moving taxes from local producers to domestic producers. This is aptly done by increasing import tariffs and reducing the volumes that are collected from taxes on international trade. From Table 5, one can see that a 1 point increase in effectively applied tariff rates can reduce the size of a country's shadow economy by 0.55 percentage points. Since most countries operate with applied tariff rates of less than 5%, there can be room to increase these while also lowering other kinds of domestic taxes. Since just over 65% of countries fall into this less than 5% category, global informality has the potential to be reduced by a decent margin if countries could frame hikes in tariff rates as attempts to supplement for other types of taxes in the name of reduced informality. This can then be offset by multinational corporations increased willingness to operate in countries where there is less informality. Thus tariff rates can be used as a way to improve trade outcomes in the future. It is, however, important to remember that FDI does increase shadow economic activity, yet this affect is relatively small.

An additional policy implication relates to agriculture and comparative advantages. Simply put, countries with both high average comparative advantages and large agricultural sectors should work to diversify away from their agricultural sectors (clearly reducing comparative advantages is not a realistic nor intelligent change). Governments can act to subsize other industry and understand that this spending can be recovered from the increased revenues that are to come with a reduced shadow economy. Countries with relatively low agricultural sectors would do well to continue increasing their comparative advantages, thus focusing on capital is a solid way to reduce informality from potential gains of trade.

A final, relatively clear policy implication is that being under sanctions creates large impacts for shadow activity. This likely holds regardless of the type of sanction and whether

or not one or both are present (they all have the same impact even in magnitude). This states that an additional negative externality of poor integration with the world economy is that local underground activity will increase, thus reduce the ability of government to handle their sanctioned state.

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